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MILITARY SPECIFICATION

MIXING PLANT, ASPHALT, BATCH-MIX, MANUAL OPERATING, PORTABLE ELECTRIC-MOTOR-DRIVEN, 120 TONS PER HOUR CAPACITY

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 4,000 pound (lb) batch-mix, manual operated, portable, high-type asphalt mixing plant of not less than 120 tons per hour (tph) capacity. The mixing plant shall be comprised of towable main component units and supplementary component parts and accessories to be transported by truck or trailer.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, and Seals.
TT-P-664 - Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 3895

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TT-V-119 - Varnish, Spar, Phenolic-Resin.
TT-V-121 - Varnish, Spar, Water-Resisting.
VV-F-800 - Fuel, Oil, Diesel.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock Corner.
PPP-B-1055 - Barrier Material, Waterproofed, Flexible.

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MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173 - Varnish, Moisture- and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-R-196 - Repair Parts, Accessories and Kits, Mechanical, Packaging of.
MIL-C-3600 - Compressors, Rotary, Power-Driven; and Compressors Reciprocating, Power-Driven; Air and Gas (Except Oxygen and Refrigerant), Packaging of.
MIL-T-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-V-13811 - Varnish, Waterproofing, Electrical, Ignition.
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-E-17555 - Electronic and Electrical Equipment Accessories, and Repair Parts, Packaging and Packing of.
MIL-B-22191 - Barrier Materials, Transport, Flexible, Heat Sealable.
MIL-T-23759 - Tanks, Asphalt Storage, Wheel Mounted.
MIL-P-46093 - Primer Coating, Synthetic (For Brake Drums).
MIL-H-52213 - Heaters, Hot Oil.
MIL-C-52950 - Crates, Wood, Open and Covered.

STANDARDS

MILITARY

MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging of.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

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DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Carrier Safety Regulations.

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards, 29 CFR 1910.

(Application for copies of DoT and DoL should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B15.1 - Mechanical Power - Transmission Apparatus.

(Application for copies should be addresssed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D5 - Test for Penetration of Bituminous Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

CONSTRUCTION INDUSTRY MANUFACTURERS ASSOCIATION (CIMA)

"Live Zone" Volume Rating for Batch Type Pugmill Mixers.

(Application for copies should be addressed to the Construction Industry Manufacturers Association, 135 S. LaSalle Street, Chicago, IL 60603.)

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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ICS 1 - General Standards for Industrial Controls and Systems.
MG 1 - Motors and Generators.
WC 3 - Rubber-Insulated Wire and Cable for the Transmission and Distribution of Electrical Energy.
WC 7 - Cross-Linked-Thermosetting-Polyethylene-Insulated Wire and Cable for the Transmission and Distribution of Electrical Energy.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description. The asphalt batch mixing plant, hereinafter called the plant shall be electric-motor-driven, portable type. The plant shall consist of all units, components, and accessories to provide a complete plant capable of mixing not less than 4,000 lb per batch of high quality asphaltic mix at a capacity of not less than 120 tph. The plant shall include, but not be limited to, the following basic units:

a. One four-bin cold-aggregate feeder of hopper and frame construction used for blending and feeding predetermined amounts of aggregate to the cold-aggregate conveyor.
b. One cold-aggregate conveyor to receive cold-aggregate from the feeder and to discharge into the drier.
c. One aggregate drier of the single-rotating-drum type to heat and dry the aggregate. An automatic temperature-fuel regulating control shall be installed on the burner of the drier to regulate the amount of fuel oil required in relationship to the preset temperature regulation.
d. One fines collector of the centrifugal cyclone-type separator with connecting duct sections and an exhaust fan to provide draft, remove products of combustion, and to collect the fines emerging from the drier.
e. One hot-aggregate elevator of the totally enclosed, centrifugal-discharge, bucket type to receive the hot dried aggregate.
f. One gradation control unit of the four-bin screening type, with hot aggregate storage bins serving the mixing unit.
g. One batch unit, consisting of the batch proportioning and mixing type, to receive aggregate and mix with bitumen, as required for various mixes, equipped with structural frame, platform, ladder, scales, guardrails, controls, and supporting leg stands.
h. One fines feeder of the hopper-and-vane, screw-conveyor, or pneumatic-injection type to meter granular mineral filler into the upper batch tower unit.
i. One complete set of jacketed and unjacketed piping and accessories necessary for interconnecting all the component units in accordance with the approved arrangement plan submitted by the suppliers.
j. One wheel mounted 10,000 gallon asphalt storage tank conforming to MIL-T-23759, with magnetic solenoid temperature control valves and regulating dials.
k. One wheel mounted 10,000 gallon fuel oil storage tank.
l. One asphalt storage transfer pump, electric-motor-driven, 480-volt (V) rating, 90 gallon per minute minimum capacity.
m. Three asphalt melters of the type normally furnished to the commercial market. Each melter shall be capable of melting not less than 750 gallons per hour of 85-100 penetration asphalt in 55 gallon drum containers.
n. Two 3-million British thermal units per hour hot oil heaters conforming to MIL-H-52213.

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o. Motor control center complete with motor controls, circuit breakers, switches, relays, transformers, and auxiliary equipment, properly wired and operable as a central electrical control for the plant.
p. Portable power and control cables of the proper size and number necessary to provide power and control circuits from the control center to each motor and control station of the plant.

The above units (except items l through p) shall be mounted on either semitrailers or cart type trailers and may be grouped at the discretion of the supplier as long as the plant can be interconnected mechanically and electrically and operated at rated capacity.

3.1.1 Capacity and performance. The output of the plant, when stepped up to high capacity, shall be capable of capacity production of high type, dense graded, bituminous concrete. The design mix shall be within the limits set forth in table I. After the design mix is established, the plant shall be capable of producing the design mix with variations not exceeding the following: Passing number 4 sieve or larger, ± 4 percent; passing number 10, 40, and 80 sieves, ± 3 percent; passing number 200 sieves, ± 1.5 percent, asphalt, ± 0.3 percent.

TABLE I. Mix envelope.

Square mesh laboratory sieve	Percentage by weight
1 inch	87-100
1/2 inch	70-79
No. 4	48-60
No. 10	37-49
No. 40	20-29
No. 80	12-19
No. 200	4.5-7.5
Asphalt	5-6.5

3.1.2 Compatability. With plant and piping installed and the electrical circuits connected, all component units of the plant shall be capable of being made ready for full production operation within 5 days, using three field mechanics for testing and adjustment.

3.1.3 Bearings. All bearings shall be antifriction type, greasable or sealed type with dust seal provision.

3.2 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection (see 4.2.1 and 6.5).

3.3 Standard commercial product. The plant shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features

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which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the plant being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Materials. Materials shall be as specified herein and in applicable specifications and standards, and other referenced documents. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials shall be free of defects which adversely affect performance or serviceability of the finished product.

3.5 Safety. All exposed parts which are subject to high operating temperatures of which are energized electrically, and all moving parts which are of such nature or so located as to become a hazard to operating or maintenance personnel, shall be enclosed or guarded. Protective devices shall not impair the operating functions. Each trailer shall conform to all applicable requirements or the DoT Federal Motor Carrier Safety Regulations. The mixing plant shall comply with all applicable OSHA requirements in effect on date of manufacturer. Reflectors are required.

3.6 Maintainability. The plant design together with component and accessory location and installation, shall permit ready accessibility to all items requiring periodic maintenance service in the field, which will be accomplished with the use of conventional, general-purpose tools associated with equipment of this nature. The replacement and adjustment of components and accessories shall be accomplished with minimum drainage requirements and minimum disturbance to other elements of the plant.

3.7 Transportability. Each wheeled component, shall be capable of travel over smooth hard-surfaced primary highways at speeds up to 20 miles per hour, travel over unimproved roads and hilly cross-country terrain, and negotiate a 90 degree (°) turn without weaving.

3.7.1 Braking ability. On each wheeled component equipped with brakes, the combined service brakes of the towing and towed vehicle shall control, decelerate, and stop the combination in conformance to DoT Federal Motor Carrier Safety Regulation requirement. When actuated, the emergency breakaway system shall produce smooth emergency brake application.

3.7.2 Ground clearance. Exclusive of the running gear, the ground clearance of each unit shall be not less than 10 inches.

3.8 Servicing and adjusting. Prior to acceptance of the plant by the Government, the contractor shall service and adjust the plant for immediate operational use as required in the operator's manual. The servicing and adjusting shall include at least the following:

a. Inflation of all tires.
b. Adjustment of brakes.

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c. Proper functioning of all lighting and electrical systems.
d. Complete lubrication with grades of lubricants recommended for ambient temperature at the delivery point.

The plant shall be conspicuously tagged to identify the lubricants and their temperature range.

3.9 Vehicular component units. The cart-type trailer and semitrailer component units shall be as specified herein.

3.9.1 Intervehicular connections.

3.9.1.1 Truck-tractor and semitrailer. The semitrailer shall be furnished with airbrake standard commercial couplings, for connecting the truck-tractor airbrake system. The coupling shall be installed on the front of the semitrailer, symmetrical about the centerline, in accordance wih SAE J702. Coupling clearance shall permit connection of the truck-tractor hoses to the semitrailer without interference with the upper fifth-wheel plate or other components. The airbrake couplings shall be protected with spring covers.

3.9.1.2 Truck and cart-type trailer. All intervehicular components shall be installed on the trailer, so that they will extend between 36 and 42 inches toward the prime mover from the center of the lunette eye. Two dummy couplings shall be furnished and fastened to the tongue for doubling back and holding the lines. The intervehicular line routing, connected and stowed, shall be such that dropping of the tailgate of the truck will not damage any lines.

3.9.2 Chassis frame. The framing members of the component units shall support the weights and maintain alignment under all conditions of transport and of plant operating as specified herein.

3.9.3 Semitrailer upper fifth-wheel plate and kingpin. The upper fifth-wheel plate shall cover a 36-inch diameter lower fifth-wheel and may extend the entire width of the gooseneck. The fifth-wheel plate and kingpin shall be compatible with a fifth-wheel height of 48 inches $\pm$ 1 inch. Both longitudinal and transverse reinforcing members shall be provided. At the front of the plate, a front coupling ramp shall be furnished. Ramps that are not flush with the front crossmember shall include at least four vertical gussets and shall be provided with holes for water drainage. A 2-inch throat diameter kingpin, conforming to SAE J700, shall be installed not less than 24 inches no more than 36 from the front of the longitudinal centerline of the semitrailer. The swing clearance rearward from the kingpin to any part of the semitrailer below the upper fifth-wheel plate shall be not less than 74 inches.

3.9.4 Cart-type trailer towing attachment. A removable towing attachment shall be attached to the cart-type trailer and shall include a lunette conforming to SAE J847. When the cart-type trailer is in the level towing position, the height of the lunette shall be between 32 and 36 inches from the ground line. Two safety chains shall be furnished on each trailer.

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3.9.5 Landing jacks and jack legs. Landing jacks shall be provided on semitrailers and jack legs on cart-type trailers for raising, leveling, and supporting each vehicular component unit, except for the hot-aggregate elevator. The landing jacks and jack legs shall be hand or power operated hydraulic units, and shall be permanently installed on the vehicular component unit. The landing jacks and jack legs shall be of such strength, design, and location that the coupling and uncoupling the trailer, with a towing vehicle, will not cause excess stresses. Operating jacks shall be furnished to raise each vehicular component unit, except the batch type minimum unit, into position for plant operation. The operating jacks shall be either permanently attached or removable. When the jacks are removable, provisions shall be made for stowing the jacks on the vehicular component unit. Operating jacks shall be of such strength, design, and location that operating of the plant under load will not cause excess or uneven stresses. Any jack may serve as a combination landing jack or jack leg, and operating jack providing it meets all requirements specified herein.

3.9.6 Running gear. The axle, or the forward axle on tandem axles, shall be in alignment with the longitudinal centerline or the vehicular component unit so that the two diagonal measurements, from the center of the lunette or kingpin to the outer center of each hub, do not differ in excess of 1/8 inch. The rear axle of tandem axle assemblies shall be in alignment with the front tandem axle so that the parallel measurements between hub centers on each side do not differ in excess of 1/8 inch. Stub axles or through axles may be furnished. The length of axle tubes or through axles may be increased during manufacture, providing the mounting pads are increased at least an equal distance to insure that the structural rigidity is equal to that of the axles. Axle assembly for the component units shall be either of the single or tandem axle type, as applicable for load distribution and weight. All units shall be equipped with suitable size wheels and tires, in accordance with the TRA recommendations, to withstand the maximum loads, stresses, travel speeds, and with adequate road clearance. Each single axle shall be designed for a load of not more than 18,000 lbs, and shall be compatible with the brake system in accordance with the loadings.

3.9.7 Brake system. Each wheel component unit shall be equipped with a service and emergency breakaway, waterproof, all air or air over hydraulic brake system complying with all applicable DoT Federal Motor Carrier Safety Regulations construction and performance requirements. The brake system shall include service and emergency line filters, an air reservoir, and an emergency relay valve. All brake system components shall be located, installed, and protected so that damage from ground obstructions and debris is prevented and that maximum efficiency and ease of maintenance is obtained.

3.10 Electric equipment.

3.10.1 Electric motors. All electric motors shall be 3 phrase, single speed, induction type, and connected for 460V, alternating current (ac), 60 Hertz operation. All motors shall be totally enclosed nonventilated or totally enclosed fan-cooled machines conforming to NEMA MG 1. Motors shall be

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suitable for the respective application in an ambient temperature of 104° Fahrenheit (°F) (40° Centigrade (C)). All motors, 75 horsepower (hp) and larger, shall be arranged for sequence starting from reduced voltage or current motor starters. The remaining motors may be across-the-line start with overload protection and controlled from the motor control center (see 3.10.2).

3.10.2 Motor control center. The motor control center shall be a free standing, outdoor type, combination motor control unit, rated for 460V, 3 phase, 3 wire, ac, conforming to NEMA ICS 1, in a type 3S enclosure. Provision shall be made for remote control stations (start-stop pushbuttons) where required. The motor control center, complete with all motor controllers, components, and accessories, shall be suitable for continuous operation under conditions as follows:

a. At ambient temperatures from +104°F (40°C) to +32°F (0°C).
b. Exposure to severe dust conditions.
c. Exposure to salt air.
d. At any altitude from mean sea level (msl) to 5,000 feet (1524 meters) above msl.

The motor control center shall have a manually operated main circuit breaker on the incoming power supply to the control center. The main circuit breaker may be either integral with the control center, or in a separate enclosure and shall have overload, ground fault, and short circuit protection. The main circuit breaker shall have a shunt trip, with remote trip pushbutton(s) located at the operator position(s), for emergency trip operation. The termination for the incoming power supply to the main circuit breaker shall have screw pressure type terminals for not less than three sets of power supply cables. One set of terminals (one per phase) shall be suitable for terminating conductors from No. 4 American Wire Gage (AWG) to No. 1 AWG; the other terminals shall be suitable for termination conductors from 300 thousand circular mils (MCM) to 600 MCM. The motor control center shall have an instruction plate permanently affixed in a conspicuous location. In addition to instructions for connecting and operating the plant, the instruction plate shall show the power supply requirement for the plant to bold type, as follows:

POWER SUPPLY REQUIRED
460 VOLTS, 3 PHASE, 3 WIRE
INPUT CAPACITY, MAXIMUM____KVA (_____(KW))
OPERATING LOAD (AT RATED CAPACITY)_____KVA(_____(KW))
STANDBY LOAD (HOT OIL HEATER, ASPHALT MELTERS, PUMPS)_____KVA(____(KW))

3.10.2.1 Motor controllers. Motor controllers (starters) shall be ac, general purpose, class A, magnetic controllers for single speed, nonjogging, squirrel-cage induction motors, conforming to NEMA ICS 1. Controllers for motors of 75 hp and larger shall be reduced voltage or current motor controllers. Controllers for motors of less than 75 hp may be full-voltage (across-the-line) controllers. In so far as consistant with plant operation the controllers shall be interlocked to provide sequence starting, with the

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larger motors to be started first, in order to reduce the required power input capacity to a minimum. The staring sequence shall be included in the connecting and operating instructions.

3.10.3 Power and control cables. All power, control and grounding cables, fittings, and accessories required to completely connect the plant for operation shall be furnished in the length required for each circuit. In addition, three sets of power cables (one three-conductor cable or three one-conductor cables each) not less than 100 feet in length, shall be furnished to connect the motor control center to the incoming power supply source. One set of power supply cables shall have adequate load capacity to supply the standby loads, such as the hot oil heating system, asphalt melters and asphalt storage transfer pumps, but the individual conductors shall be not smaller than No. 4 AWG. The other sets of power supply cables shall have adequate load capacity to supply the maximum power input required to operate the complete plant, but the individual conductors shall be not smaller than 400 MCM copper. All power and control cables, for the power supply and for interconnecting the plant, shall be heavy-duty portable cables, conforming to NEMA WC 3 or NEMA WC 7 and suitable for direct burial. All portable power cables shall be rated for conductor operating teperature of not less than 75°C. The load on any conductor shall not exceed the allowable ampacities of NFPA No. 70 for direct burial conductors. The grounding system for the complete plant shall include the grounding conductors, fittings, ground rods, and clamps. The grounding conductors shall be of suitable size and type for the application.

3.11 Drive components.

3.11.1 V-belts drivers. V-belts drives shall be in accordance with ANSI B15.1 and shall have provisions for adjustment.

3.11.2 Gears and gearboxes. All gears except pugmill mixer timing gears, shall be enclosed in gear boxes and shall operate in an oil bath.

3.12 Cold-aggregate feeder(s). The cold-aggregate feeder(s) shall be of hopper and frame construction and shall consist of a two-bin coarse aggregate hopper and a two-bin fine aggregate hopper, with a total capacity of not less than 100 tons, belt-conveyor feeder(s), all supported by a rigid steel frame and mounted on a pneumatic-tired semitrailer chassis. The cold-aggregate feeder(s) shall discharge the materials onto the cold-aggregate conveyor. Bin vibrators shall be provided for each compartment.

3.12.1 Hoppers. The coarse aggregate and fine aggregate hoppers shall be of welded steel construction, each divided into two bins to permit handling of two sizes of aggregate in each hopper (total of four sizes of aggregate). The bin side plates and divider plates shall prevent bridging or clogging of the aggregate. Each bin shall have an opening not less than 10 feet in width, for receiving the amount of aggregate being loaded into each hopper. Hopper and bin extensions shall be provided for both hoppers. Bins and bin extensions shall be provided with reinforcing to prevent damage to the edges while bins

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are being charged. Each bin shall be provided with an individual, adjustable, control gate, or by other suitable means, to regulate the flow of aggregate. The gate shall be provided with an adjustment lever and position locking device when control gate is used for regulating aggregate flow.

3.12.2 Belt-conveyor feeder(s). The belt-conveyor feeder(s) shall consist of a suitable steel frame, conveyor belt, head-end (drive) pulley, foot-end pulley, carrier rolls, and return rolls if required. Straight carrier rolls shall be mounted under the belt to support the aggregate as it is discharged from the aggregate hoppers. Steel side plated and rubber flashing shall be provided to prevent spillage of aggregate from the sides as it is discharged from the hopper. Pulleys and rollers shall be mounted on steel shafts, and supported by antifriction or bronze bearings with dustproof seals. Self-aligning bearings for head and tail end pulley shafts shall be furnished and shall be provided with labyrinth or lip type seals. The belt-conveyor feeder(s) foot-end section shall be provided with a takeup adjustment mechanism to obtain belt tension.

3.13 Cold-aggregate conveyor. Cold-aggregate conveyor shall be of the belt conveyor drive type of not less than 24 inches in width and manufacturer's standard length, with masts or arms for elevating the frame, a two-wheel chassis, and electric motor. The discharge capacity shall be not less than 120 tph. The conveyor frame shall be raised either hydraulically or by mechanical means. The chassis shall be constructed so that the entire unit can be towed with a drawbar.

3.14 Aggregate drier. The aggregate drier shall consist of a single rotating drum with feed or charging chute, oil burner, force draft blower and motor, fuel pump, accessories and instruments, and all other components necessary for the operation of the drier as specified herein. When drying aggregate at a discharge temperature of 290°F to 310°F, the capacity of the drier shall meet the production output rate as specified in table II.

TABLE II. Output capacity.

Aggregate	Passing square mesh screen (percent) 1 1/2 inch	No. 4	Initial moisture content (percent)	Capacity tph of production output (minimum)
Limestone	100	60	*6 or less	120
Gravel	100	60	*6 or less	120

*With not more than 1/2 percent of allowable moisture content retained after being dried.

3.14.1 Drum. The drier drum shall be a single cylinder constructed of heavy-gage steel plate, capable of withstanding the abrasion, wear and heat distortion at peak production for long usage without replacement. The drum shall be provided with replaceable lifting flights which shall lift and drop the aggregate in a thin veil through the hot gases produced by the burner. Suitable means shall be provided to discharge the dried aggregate into the

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boot of the hot-aggregate elevator. The drum shall be fitted with steel supporting tires attached to the outer circumference. Each tire shall revolve on not less than two adjustable trunnions made of hardened cast steel. Means shall be provided for controlling the longitudinal movement of the drum along the longitudinal axis of the drum in both the uphill and downhill direction. The drum shall be driven by a positive or friction drive arrangement. All bearings used in the driving mechanism shall be of the antifriction type. The charging end of the drier drum shall include a flanged exhaust gas duct connection onto which the fines collector duct shall be attached.

3.14.2 Drier burner and fuel system. The drier fuel system shall include burner, fuel piping, fittings, valves, fuel pump, strainers, sediment and moisture bowl, relief valve, pressure gages, and other items which may be required to provide a complete operating drier fuel system.

3.14.2.1 Burner. The drier burner shall be low pressure, air atomizing, and oil burning. The burner shall meet the performance requirements specified herein, when operated with fuel oil conforming to VV-F-800, grade DF-A, DF-1, or DF-2, and jet fuel conforming to MIL-T-5624, grade JP-4 or JP-5. The burner shall have the capacity of drying the wet aggregates to meet the specified production requirement, and shall operate in conjunction with the forced draft blower for efficient combustion.

3.14.2.2 Fuel piping and pipe fittings. Fuel pipe shall be black iron and pipe fittings shall be malleable iron. Unions shall be installed, where required, for disassembly of piping, valves, and accessories. On the discharge line of the fuel oil pump, a tee-off line with stop valve shall be provided for refilling the fuel tanks of auxiliary generator sets.

3.14.2.3 Fuel control device. A control device shall be installed in the fuel supply system to the burner in a location accessible to the operator. The control device shall provide an infinitely variable range of adjustment and shall maintain a steady flow of fuel at any setting. The control device shall include a calibrated dial with pointer to show flowsetting. A gate type, shut off valve shall be installed in the fuel supply line to the burner. The relief valve shall have a capacity not less than the capacity of the fuel pump. A fuel tank return line shall be connected to the discharge of the relief valve drains to permit complete drainage of the piping.

3.14.2.4 Fuel pump. The fuel pump shall be of the positive displacement type for pumping the required fuels to supply the drier heat demand. This capacity shall be developed when pumping against a discharge pressure not less than the combined friction loss created by the burner and fuel supply system. The pump shall be self-priming, with an electric motor drive, and shall be permanently mounted in an accessible location.

3.14.2.5 Strainer. An oil strainer shall be provided and installed in the fuel supply line in such a manner as to permit easy accessibility for cleaning. The strainer shall have the capacity to remove all foreign matter which might foul the burner.

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3.14.2.6 Advance temperature detector. An advance temperature detecting device shall be provided for compensating the sudden surge of fuel temperature relationship demand in the drier when charged with large load of aggregate with higher or lower than normal moisture content. The device shall be capable of overriding the set fuel control unit until temperatures are established.

3.14.2.7 Pressure gage. An oil pressure gage shall be installed in the burner fuel supply system. The gage shall operate with the fuel oils specified for the fuel pump, shall have a heavy-gage metal case and an indicating dial with a range of 0 to 100 pound-force per square inch gage, and shall be provided with a shutoff cock or valve. The dial shall be graduated in increments of not less than 5 pound-force per square inch (psi).

3.14.2.8 Temperature indicator. A temperatue indicator shall be provided to indicate the temperature of the aggregate as it is discharged from the drier. The indicator shall have a range of 100°F to 500°F.

3.14.2.9 Level indicator. A level indicator shall be provided and shall be mounted on the drier assembly in such a manner as to indicate the angle of the longitudinal axis of the drier drum to the horizontal. The indicator shall include a graduated scale in increments not greater than 1 and with a range of 0° to 10°. The indicator shall be mounted in a location where it is visible.

3.14.2.10 Driving mechanism. All necessary gearboxes, shafting, motor, chains, bearings, and any components required for a complete operating drive system shall be provided.

3.14.2.11 Gearboxes. The main driven shall be provided with a gearbox to the power source. Gears shall run in an oil bath. Gearboxes shall be provided with oil seals to prevent loss of oil and entry of foreign material.

3.14.2.12 Forced draft blower. A centrifugal type blower shall be provided to supply air for atomization to the drier burner. The forced draft blower shall have a centrifugal type wheel, steel shaft and housing, and drive. The forced draft blower shall have a capacity to provide a sufficient quantity of air to the burner to assure efficient operation of the burner at maximum burner capacity. The forced draft blower shall be provided with antifriction type bearings, and the intake shall be provided with an expanded metal or wire mesh screen.

3.14.3 Discharge elevator and discharge chute. Means shall be provided at the burner end of the drier drum to discharge the dried aggregate into a discharge chute which will, in turn, direct the aggregate into the boot of the hot-aggregate elevator. The elevator discharge shall be of such a height as to eliminate the need for a hot-aggregate elevator pit.

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3.15 Fines collector. The fines collector shall consist of a motor, an exhaust fan, steel stack, centrifugal type cyclone collector(s), screw conveyor, metal duct, all necessary drives, gearboxes, controls, instruments, accessories, and other components necessary for the operation of the unit.

3.15.1 Exhaust fan. The exhaust fan shall be of a type for the intended application and shall provide sufficient draft through the drier and fines collector to result in an efficient separator operation. The exhaust fan shall have a single inlet to which the duct from the fines collector shall be attached and a flanged discharge for connection of the stack. The fan shall be provided with a structural steel base which shall be bolted to the semitrailer chassis. The base and fan housing shall rigidly support the weight of the steel stack. The fan housing shall be provided with inspection and cleanout plates.

3.15.2 Stack. A removable stack, of a size compatible with the exhaust fan, shall be provided and shall be flange mounted on the discharge of the exhaust fan. The stack shall be constructed of sheet steel.

3.15.3 Collector. The collector shall be of the centrifugal cyclone type fabricated of welded sheet steel, cast steel, or cast iron, and be complete with all necessary flanges for inlet and outlet connections. The capacity shall be such as to be compatible with the exhaust fan.

3.15.4 Collector conveyor. A screw or enclosed inlet conveyor shall be attached to the bottom of the fines collector and shall convey the fines from the collector to the boot of the hot-aggregate elevator. The conveyor tube shall be provided with inspection plates and cleanouts and with a discharge opening at the end. The conveyor may be constructed in sections securely bolted together.

3.15.5 Duct work. All metal duct work and fitting required for connecting the exhaust gas outlet of the drier to the inlet of the fines collector and to connect the outlet from the fines collector or the exhaust fan shall be provided. All necessary draw bands, gaskets, and flanged connections shall provide a tight duct system, without excess stresses, when assembled for operation. Ducts shall be provided with structural steel supports or shall be self-supporting to assure a sturdy installation. All ducts shall be removable for towing purpose. A damper shall be provided for manually regulating the draft.

3.16 Dust collector (bag house). When specified (see 6.2), a fabric filler type dust collector system shall be furnished. The system shall be designed to be used in conjunction with the plant fines collector system. The collector shall be the manufacturer's standard commercial design and shall be sized for the capacity of the plant specified. The collector shall be portable type mounted on a semitrailer. The collector shall be completely assembled and shall include all components and accessories normally furnished with the system including, but not limited to the following:

a. Collector bag housing.
b. Dust collection hoppers.

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c. Inlet and exhaust system.
d. Compressed air system.
e. Diesel power unit with generator.
f. Safety and operating controls.

3.17 Hot-aggregate elevator. A hot-aggregate elevator of the centrifugal discharged bucket type, totally enclosed, shall be provided. The elevator shall be compatible in height with the upper batch tower unit and shall have a capacity equal to, or greater than, that of the vibrating screen unit. The elevator shall be totally enclosed, provided with intake ports, mounted on pneumatic tires, and shall have towing attachments.

3.18 Gradation control unit. The gradation control unit shall consist of a vibrating screen unit with motor and vibrating mechanism, a four-compartment bin of not less than 24 tons in total working capacity, with control gates, bin level indicators, chutes, housings, ladders, platforms, guards, and all other components required for the operation of the unit as specified herein.

3.19 Capacity. The gradation control unit shall have a capacity to screen and feed not less than 120 tph of the dried limestone or gravel aggregates in accordance with table I.

3.19.1 Vibrating screen. A horizontal or inclined vibrating screen assembly, with interchangeable wire screen cloths of the sizes specified herein, shall be provided. The vibrating screen assembly shall grade the aggregate into four uniform sizes and scalp of oversize material. The screen shall discharge each of the four sizes of aggregate into a separate compartment with the smallest aggregate being discharged into the largest (fine sand) compartment. The top deck shall be provided with an overflow chute to scalp off oversize aggregate. The screen shall be equipped with screen cloths of the following sizes: 2-, 1-, 3/4-, 1/2-, 3/8-, 1/4-, No. 4 and No. 10, all with square openings. The screen shall consist of a stationary frame upon which is mounted a frame with the specified cloths and with a vibrating mechanism. Three screen cloths, of sizes specified for each frame, shall be provided with the vibrating screens.

3.19.2 Stationary frame. The stationary frame of the screen shall be consructed of structural steel members or steel castings cross braced to prevent distortion under maximum loads and operating conditions as specified herein.

3.19.2.1 Live frame. The live frame of the screeen assembly shall be constructed of steel plated or structural steel members of rigid construction, with sufficient crossmembers to securely hold the width to insure fitting of screen cloths. The live frame shall be provided with means for holding the replaceable screen cloth securely in place to prevent movement of the cloth during full-load operation. A series of longitudinal bars shall be provided to support the screen cloth. Provisions shall be made for changing cloth by using common tools. The frame shall be balanced to provide an even flow of material across the full width of the cloth.

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3.19.2.2 Screen cloth. All screen cloth shall be made of crimping grade, oil-tempered wire, carbon range 0.55 to 0.70, drawn and hardened. Size of wire for 3/8-inch screen openings and above shall be of the standard medium type; for 1/4-inch screen opening and below shall be standard light type. Cloth size openings shall be equal to US Test Seive of the mesh specified.

3.19.2.3 Vibrating mechanism. A mechanical vibrating mechanism shall be provided for activating the live frame assembly. If material from the deck or decks of the screen impinges on the activating mechanism, the mechanism shall be equipped with a replaceable, protective cover.

3.19.2.4 Screen housing. The entire screen assembly shall be housed with heavy-gage sheet steel to retain dust and heat. Access openings and covers shall be provided at all points requiring inspection or maintenance.

3.19.3 Bin. The bin shall be fabricated of heavy-gage sheet steel braced and reinforced with structural steel shapes. The bin shall have a working capacity of not less than 24 tons of aggregate and shall be divided into four compartments capable of meeting the design mix at the plant capacity. Sampling device shall be provided for each bin compartment.

3.19.3.1 Telltales and overflows. Each of the four compartments shall be provided within overflow chute and telltales to indicate, and audibly warn, when the level of aggregate within the bin is insufficient for mixer operation. The audible warning may be a horn, bell, or similar device.

3.19.3.2 Sampling. Means shall be provided for obtaining representative samples from each bin compartment. All accessories necessary for obtaining samples shall be furnished.

3.19.3.3 Walkways and ladders. Steel walkways shall be provided at the top of the bin for access to bin compartments and screens for inspection, servicing, and maintenance. The walkways shall be provided with safety tread, handrails, and ladders.

3.19.3.4 Fines feeder. The fines feeder shall consist of a pneumatic injection system or a hopper and screw conveyor, drives, supporting frame, and all accessories or other components required for the operation of the fines feeder as specified herein. The fines feeder shall be either ground mounted or integrally mounted with other components.

3.20 Batch unit. The batch unit shall consist of a hopper, transfer pump batching system, spray bar or spreader bucket, pugmill mixer, discharge chutes, and all other accessories piping, and components necessary for the operation of the unit. The batch unit shall be supported to provide a minimum height of 10 feet between the discharge gate and ground level.

3.20.1 Capacity. The batch unit shall have a capacity of mixing not less than 4,000 lb per batch.

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3.20.2 Transfer pump. The transfer pump shall be the positive displacement type, equipped with a pressure regulating device, all necessary piping, and an all-iron body jacketed with hot oil for pumping the bitumen to the batching system from an external source. The pump shall have a capacity compatible to the plant capacity for transferring the bitumen of penetration grade 85 to 100 heated to a temperature of 250°F to 320°F. Penetration grades shall be determined by ASTM D5. The pump shall be provided with control located on the operator's station.

3.20.3 Measuring and injection system. A measuring and batching system or the volumetric or weight bucket type, shall be provided to supply bitumen to the spray bars or spreader bucket. The system shall have an accuracy of within ± 1 percent, when using bitumen with a penetration grade of 85 to 100, at a temperature between 250°F and 325°F.

3.20.4 Pugmill mixer. The pugmill mixer shall consist of an enclosed steel chamber having not less than 50 cubic feet rated live zone capacity when figured under formula adopted by CIMA for "Live Zone" Volume Rating of Batch Type Pugmill Mixers. The size of the pugmill mixer shall be such as to produce the capacities specified in 3.1.1. Paddle arm arrangement shall be such that any one paddle or wearing sleeve arm may be removed or replaced without disturbing any of the other paddle arms. The shafts shall be constructed of solid steel and the paddle arms of high carbon or alloy steel. The paddle tips shall be replaceable, or be covered with a replaceable wearing sleeve, and be constructed of abrasive-resisting cast iron or steel. The pugmill mixer shall be lined with replaceable abrasive-resistant liner. The pugmill mixer shall be jacketed for heating with hot oil. The paddle shafts shall be supported at each end with steel supports and antifriction bearings located outside of the mixing chamber. The pugmill housing shall be constructed of steel and shall have a steel inspection cover. The pugmill shall be provided with the necessary drives or gear trains to receive power from the motor for driving the mixer. The mixer shall be provided with control and timers located at the operator's station. Two adjustable timers shall be provided; one for wet mix, and one for dry mix. Mixer discharge gate shall be either hydraulically or pneumatically operated for unloading.

3.20.5 Controls and instrument panel. All controls such as weighing scales, thermometers, gages, indicating light signals, levers, drier burner controls, and other accessories for the operation of mixer and gradation screening units, shall be located on the operator's station. When specified (see 6.2), an automatic batching control system shall be furnished. The system shall include dry and wet mix timers, and automatic cycling device. Provisions shall be made for switching the system to manual operation.

3.21 Piping and accessories. All piping and accessories, which are required for interconnecting of component units to form a complete operating plant, shall be furnished by the contractor in accordance with the arrangement plan of the plant approved by the government. Where sectionalization is required on the hot oil asphalt lines for assembling and disassembling of the plant, flange connections with fiber gaskets shall be installed. Wire braided

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flexible hoses shall be provided for the hot oil and asphalt lines for connecting to the hot oil heaters, melters, asphalt storage transfer pump, and the asphalt storage tanks or the pugmill mixer, if required. A jacketed recirculating hot oil asphalt line from the asphalt weigh bucket or hopper to the 10,000 gallon asphalt tank shall be provided for normal plant layout. All piping, fittings, valves, and other accessories to make a complete plant shall be of steel, iron body, or malleable iron. Drier burner fuel piping shall be capable of withstanding 100 psi hydrostatic test pressure; hot oil piping, including jackets, 100 psi; asphalt piping, 50 psi.

3.22 Lubrication. Means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, 1,000 psi or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.23 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.24 Lifting and tiedown attachments. When specified (see 6.2), the component units of the plant shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform to type II or type III of MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically attached to the plant. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments may be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tie down of the plant on the carrier when shipped.

3.25 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to

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a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.26 Data plates.

3.26.1 Instruction plates. The component units which require operational instruction shall be equipped with instruction plates, including warnings and cautions suitably located and permanently affixed, describing an special or important procedure to be followed in operating and servicing the plant, inculding complete lubricating instruction. All controls shall be identified indicating the direction of motion and the operation to be performed. This may be accomplished by the use of one or more plates. Plates shall be manufactured of noncorrosive, durable material and shall be permanently secured on the plant.

3.26.2 Identification plate. An identification plate will be furnished by the contracting officer for each component unit . The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose, and securely affix a plate to each component unit in a conspicuous place with nonferrous screws, rivets, or bolts not less than 1/8-inch in diameter. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

3.27 Special maintenance tools. Any special tools required for field maintenance and operation of the equipment shall be packed in the toolbox and shipped with the equipment. Special tools are defined as those not normally included in a mechanic's tool kit, but which are designed specifically for a certain purpose on a specific piece of equipment. Means shall be provided to carry those tools and accessories required to operate the unit. Toolboxes, when furnished to hold tools and accessories, shall be made of sheet steel 0.0747-inch (US revised standard gage No. 14) nominal thickness.

3.28 Workmanship.

3.28.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.28.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

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3.28.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.28.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.28.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

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4.2.1 First article inspection. The first article inspection shall be performed on one plant when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.2 and the tests of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.2, the tests of 4.4, and the packaging inspection of 4.5. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Examination. Each plant shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests. The component units shall be arranged in the order as shown in the approved arrangements plan of the plant and the piping and accessories shall be connected to the component units for compatibility and subjected to test in accordance with 4.4.1 thrugh 4.4.5. Nonconformance to the requirements as specified in section 3 shall constitute failure and cause for rejection.

4.4.1 Test conditions. Prior to being tested, the plant shall be lubricated. Oils shall be those designated for use in the ambient temperature at which the tests are being performed. During all plant tests, the motors shall be run at normal operating speeds. Where individual components are tested with the use of power sources other than the plant power supply, the component will be operated at the normal plant operating speeds. As services and adjustments are performed, the plant shall be checked for ease of maintenance.

4.4.2 Compatibility. Assemble and operate all components units in the arrangement as shown in the approved arrangement plan. The test may be conducted with load and without load for at least 1/2 hour, provided that operating with or without loads will cause no damage to any unit or to the plant. All component units shall assemble and operate compatibility without modifications. The piping, hose, valves, fittings, and accessories shall be assembled, and matchmarks shall be painted on each of the piping, fittings, valves, and accessories, before disassembled for shipment, for the purpose of easy identification when reassembled in the field.

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4.4.3 Drying test. Unless otherwise proven by previous test records for the drying capacity, the drier shall be tested to demonstrate the 120 tph capacity of drying aggregate having 6 percent free moisture.

4.4.4 Vehicular component units performance test. The component units equipped with axles and rubber tired wheels shall be towed by a wheeled tractor or truck tractor, over unpaved roads, for not less than 10 miles, then operate cross-country for not less than 1/2 mile. During or after the test on each trailer, trailing, turning, and braking ability shall be observed.

4.4.5 Lifting and tying down attachment tests. When lifting and tying down attachments are required, they shall be tested to conform to 3.24.

4.5 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly and matchmarking. Disassembly shall be the minimum necessary to safeguard parts known to be subject to damage or loss, and to accomplish reduction in cube. Bolts, nuts, screws, pins, and washers removed shall be reinstalled in one of the mating parts and secured to prevent their loss. When necessary, to facilitate reassembly, parts removed and mating parts shall be matchmarked. Large sections and components shall be matchmarked with lusterless white enamel letter. Matchmarking shall be overcoated with clear spar varnish conforming to TT-V-121. Small parts that are removed shall have matchmarking information marked on weatherproof tags attached to mating parts. Waterproof ink shall be used for marking of information on tags.

5.1.1.2 Methods of preservation. Cleaning process, drying procedures, preservatives, and methods of preservation are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specification.

5.1.1.3 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

5.1.1.4 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 and not specifically provided for herein shall be preserved as follows:

5.1.1.4.1 Unfinished (not mechanical) surfaces. Unfinished exterior metal surfaces shall be coated with type P-1 preservative.

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5.1.1.4.2 Mechanical surfaces. Exposed mechanical surfaces shall be coated with type P-6 or P-11 preservative and wrapped or covered, as applicable, with barrier material conforming to MIL-B-121, type I, grade A, class 2 or MIL-B-22191, type II. The material shall be secured in place with waterproof tape.

5.1.1.5 Gears.

5.1.1.5.1 Exposed gears. Exposed gears shall be coated with type P-1 preservative, or a thin film of primer conforming to TT-P-664 or MIL-P-46093.

5.1.1.5.2 Enclosed gears. Gear housings containing lubricant conforming to the applicable lubricating order shall be checked for operating level and the gears actuated to insure coating of all interior surfaces.

5.1.1.6 Drive chains:

5.1.1.6.1 Enclosed chains. Enclosed chains and chain housings shall be preserved as specified for enclosed gears.

5.1.1.6.2 Exposed chains. Exposed chains shall be coated with enough P-3 or P-9 preservative to assure penetration of the preservative to the inner surfaces of the rollers, pins, and bushings. After the excess preservative has drained, the chains and sprockets shall be coated with type P-1 preservative.

5.1.1.7 Brakes.

5.1.1.7.1 Brake drum assemblies. All metal surfaces enclosed within brake drums and the braking surface or face of the brake drums shall be coated with a thin film of primer conforming to TT-P-664 or MIL-P-46093. Care shall be exercised to prevent the primer from coming in contact with brake lining, rubber, and rubber impregnated parts.

5.1.1.7.2 Air actuated brake systems. The air actuated brake systems shall be drained of condensation. Interior surfaces of the air supply tanks shall be fogged with type P-9 or P-10 preservative. Excess preservative shall be allowed to drain and the drain plugs reinstalled after preserving the threads with P-10. Air emergency quick release and relay valves not equipped with exhaust check valves shall be closed by inserting pipe plugs or by sealing the ports with waterproof. Waterproof tags shall be attached to the valves and serviceline indicating that: "The plugs or tape must be removed from the exhaust ports before placing the equipment in operation". Waterproof ink shall be used for marking of information on tags.

5.1.1.8 Hydraulic systems. The hydraulic fluid supply tank shall be filled to the operating level with approved hydraulic fluid. Where practicable, pistons shall be in their fully retracted positions, with any remaining unpainted surfaces coated with P-1 preservative. Where pistons

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cannot be fully retracted, exposed areas of normally retractable surfaces shall be coated with type P-6 or P-11 preservative. The type P-6 or P-11 preservative coated surfaces shall be spirally wrapped with barrier material conforming to MIL-B-121, type II, grade A, class 2. The wraps shall be secured in place with waterproof tape. All other unpainted exterior surfaces shall be coated with type P-1 preservative. All assemblies actuated by the hydraulic system, shall be secured to prevent movement.

5.1.1.9 Conveyor belts. Metal belt fasteners shall be coated with water resisting varnish conforming to TT-V-119 or MIL-V-13811. The belts on the wheel mounted conveyor shall be removed. Belts on other units shall be removed or released from tension. Removed belts shall be compactly rolled. Prior to rolling, belt surfaces shall be dusted with powdered talc to prevent surfaces from adhering together. When required to hold belt in compact roll, flat steel strapping conforming to QQ-S-781, class 1, type I, finish B, size as applicable shall be placed around the roll. Suitable protective cushioning shall be placed under the strapping and the strapping shall not be tensioned so tight as to damage the belting. Each rolled belt shall be wrapped with waterproofed barrier conforming to PPP-B-1055, class E-2 or H-1, with all laps, seams, and folds sealed with a water adhesive or tape.

5.1.1.10 V-belts, flat belts, and pulleys. Unpainted surfaces of pulleys shall be coated with primer conforming to TT-P-664 or MIL-P-46093. The belts for each unit shall be preserved method IC-I or IC-3, large flat belts shall be preserved and packaged as specified for conveyor belts.

5.1.1.11 Drum and rotary elevators. All exterior unpainted metal of the drier drum, rotary elevator, thrust rollers, sprockets, and unpainted metal surfaces within the drum and rotary elevators, shall be coated with type P-1 preservative. Buckets and carrying chains which are subject to movement and damage to the housing of the elevator during shipment shall be removed. Th chain shall be coiled and secured with evenly spaced wire tires. The removed buckets and chains shall be consolidated in boxes as specified in 5.1.1.24.

5.1.1.12 Pump and burners. The interior surfaces of the pump, piping valves, jets, nozzles, and burners shall be coated with type P-10 preservative in a manner to insure thorough coating of all interior surfaces. Excess preservative shall be drained. All openings shall be sealed with metal or plastic caps or plugs, or sealed with waterproof tape.

5.1.1.13 Fuel tanks. The interior surfaces of fuel tanks, if uncoated or not fabricated of corrosion-resistant material, shall be fogged with type P-10 preservative. Excess preservative shall be drained and the threads of the drain plugs shall be coated with P-10 before reinstalling.

5.1.1.14 Pugmill, bunker feeder, dust collector, and shaker. Exterior unpainted metal surfaces of the units shall be coated with type P-1 preservative. Interior unpainted ferrous metal surfaces of the mixing chamber, bunkers including shafts, screens and paddles, shall be coated with type P-1 preservative.

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5.1.1.15 Electrical components. Electrical motors shall be preserved level A by the alternate requirements of MIL-E-16298. Other electrical components shall be preserved level A in accordance with MIL-E-17555.

5.1.1.16 Air compressors. The air compressors, parts, and accessories shall be preserved in accordance with MIL-C-3600, level A.

5.1.1.17 Wire rope. The wire rope shall be coated with a preservative and secured to prevent movement.

5.1.1.18 Openings. Openings in the equipment which will trap free falling water shall be sealed by any suitable means to prevent the entry of water.

5.1.1.19 Piping equipment. Loose piping equipment required to complete the water-supply system shall be preserved method I, or III as applicable. Exposed threaded surface shall be protected against damage by the use of plastic thread protectors.

5.1.1.20 Lubrication. Lubricate all grease fittings with type P-11 preservative to include universal joints, chassis suspension and steering linkage, fifth wheels, propeller shaft, and parking brake cable. Wheel bearings requiring a NLGI No. 2 grade shall be lubricated with type P-11 preservative. Lubricate control mechanisms, linkage, hinges, fasteners, and all pivot points with P-10.

5.1.1.21 Rubber tires. Tires mounted on the wheels (road tires) of the vehicle shall be inflated to 10 lbs above the specified operating pressure.

5.1.1.22 Repair parts. Unless othewise spsecified, the preservative application criteria and applicable methods of preservation of MIL-P-116 shall be used to preserve repair parts. When specified (see 6.2), the repair parts shall be preserved in accordance with level A requirements of MIL-R-196, or when parts are not specifically covered in MIL-R-196, an applicable submethod of preservation of MIL-P-116 shall be used.

5.1.1.23 Technical publications. Technical publications for each piece of equipment shall be preserved method IC-1 or IC-3.

5.1.1.24 Consolidation. The tool boxes shall be utilized for the consolidation of loose components, tools, and publications. The remainder of the components that will not fit in tool boxes shall be consolidated, along with disassembled components, in boxes conforming to PPP-B-601, military overseas type or PPP-B-621, class 2. Boxes shall be placed in the cab or secured to the equipment with appropriate strapping. Arrangement and location on the equipment shall be such so as not to increase cubage or interfere with lifting or mobility of the equipment.

5.1.2 Level C. The complete asphalt plant shall be preserved and packaged in a manner that will provide protection against deterioration and damage from the supplier to the initial destination. This level may conform to the supplier's standard practice when such meets the requirements of this level.

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5.2 Packing. The packaging shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Trailer mounted units (except wheel mounted conveyors). Trailer mounted units shall be shipped mobile and uncrated. Movable components shall be blocked and braced so needed to prevent movement during shipment.

5.2.1.2 Wheel mounted conveyors. The conveyor shall be disassembled. The conveyor belt and electric motor shall be packed in boxes conforming to PPP-B-601, military overseas type or PPP-B-621, class 2. The wheels and chassis components shall be packed in open type crates conforming to MIL-C-52950, style A or B, type optional. Sections of the conveyor shall be shipped uncrated or bundled. When bundling is used the sections shall be nested, arranged, and secured with suitable wood blocking and battens, as required to form compact nonshifting bundles. Strapping shall be 0.035 by 1 1/4-inch flat steel conforming to QQ-S-781, class 1, type I, finish B. The bottom of the secured load shall be fitted with wooden skids extending the full width of the load to accommodate forks and slings for handling.

5.2.1.3 Components not trailer mounted. Components not trailer mounted shall be packed in accordance with the level A requirements of MIL-STD-794. Containers shall be selected from table II of MIL-STD-794 for level A requirements.

5.2.1.4 Piping and accessories. Piping suitable for bundling shall be bundled in secured lifts. The piping shall be nested, arranged, and secured with bolts or steel straps, or a combination of both, to form compact nonshifting bundles. Strapping shall be 0.023 by 3/4-inch flat steel strapping conforming to QQ-S-781, class 1, type I, finish B. Strapping shall be spaced not to exceed approximately 36 inches on center, with end straps not more than 18 inches from each end. Metal edge protectors shall be used under strapping when strapping bears on the pipe. The bundles shall be packed in open type crates conforming to MIL-C-52950, style A or B, type V. Short pieces of pipe, hose fittings, and accessories shall be packed in close-fitting boxes conforming to PPP-B-621, class 2 or PPP-B-601, military overseas type. The contents in the boxes and crates shall be cushioned, blocked, and braced to prevent movement.

5.2.2 Level B. Level B requirements shall be the same as level A except that components not trailer mounted shall be packed in containers designated for level A from table II of MIL-STD-794.

5.2.3 Level C. The complete plant shall be prepared for shipment in a manner which will insure arrival at destination in satisfactory condition. Loose components that are subject to damage during shipment shall be blocked, braced, or otherwise secured to prevent movement and damage during shipment. The toolbox lids shall be closed and secured in a manner to prevent pilferage. When sufficient space is available, packaged and unpackaged

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components required for each unit or sections shall be placed and secured on the unit or sections for which lifting or mobility of the units. Boxed and crated components shall be shipped the same time as the mobile units. Packing and containers shall comply with applicable carrier rules and regulations.

6. NOTES

6.1 Intended use. The completely equipped plant is intended primarily for use in asphaltic paving of airfield, road, street, parking lot, large floor area surfaces, or other uses.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When specifications and standards are other than as specified (see 2.1.1).
c. When first article is required for inspection and approval (see 3.2, 4.2.1, and 6.5).
d. When dust collector (bag house) is required (see 3.16).
e. When automatic batching control system shall be furnished (see 3.20.5).
f. When fungus resistance is required (see 3.23).
g. When lifting and tying down attachments are required (see 3.24).
h. Color of finish coat required (see 3.25).
i. Level of preservation and level of packing required (see 5.1 and 5.2).
j. When repair parts are to be preserved in accordance with level A of MIL-R-196 or MIL-P-116 (see 5.1.1.22).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of DoD Federal Acquisition Regulations (FAR) Supplement 27.410-6, the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DoD FAR 27.410-6 are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 Electrical power requirements. The electrical power requirements of the plant are compatible with the electrical power available from tactical utility generator sets in the DoD Mobile Electrial Power Engine Generator Set Family. After power requirements of the plant have been determined, the model(s) of generator set(s) required may be selected from MIL-STD-633, and obtained in accordance with applicable departmental policy.

6.5 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

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6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.7 Subject term (key word) listing.

Asphalt, plant, batch-type, semitrailer mounted
Mixer, pugmill, asphalt
Semitrailer, asphalt plant mounted

Custodian:
Navy - YD

Preparing Activity:
Navy - YD

(Project 3895-N233)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-29157B(YD)	MIXING PLANT, ASPHALT, BATCH-MIX, MANUAL OPERATING, PORTABLE ELECTRIC-MOTOR-DRIVEN, 120 TONS PER HOUR CAPACITY

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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